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Managed Databases

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Recap – Databases (DBs)

A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS). Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.

- Oracle (https://www.oracle.com/database/what-is-database/)

*A **database**, often abbreviated as DB, is a collection of information organized in such a way that a computer program can quickly select desired pieces of data.*

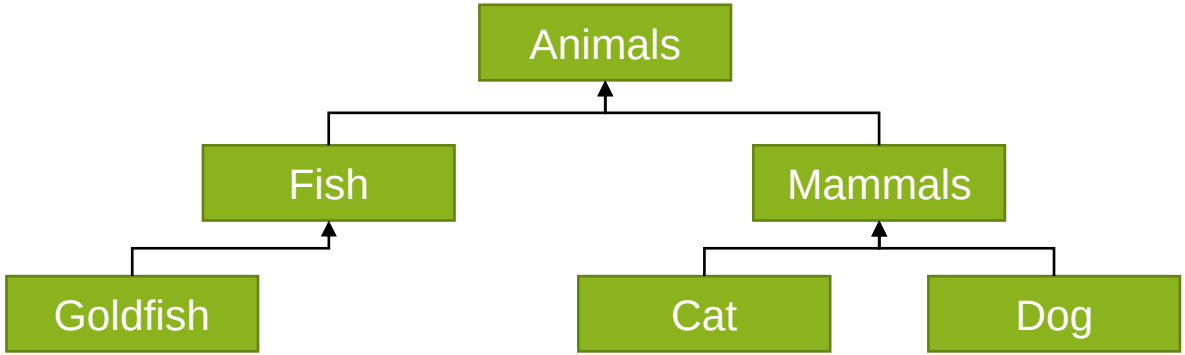
- Vangie Beal (https://www.webopedia.com/definitions/database/)

Recap – DB Types

- Hierarchical DBs
- Network DBs
- Relational DBs
- Non-relational DBs (NoSQL)

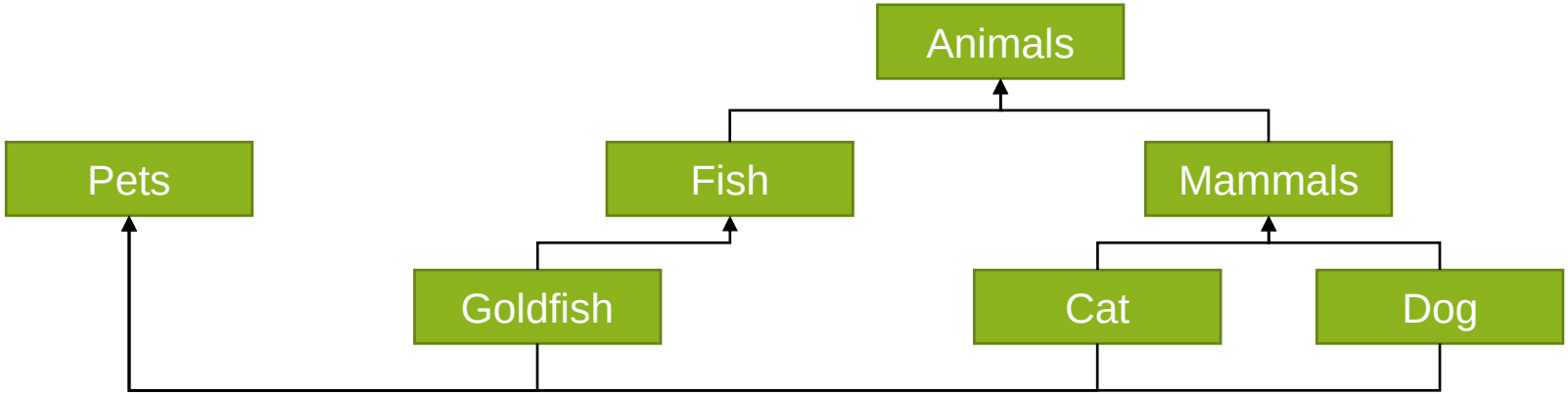
Recap – Hierarchical DBs

- Data is stored in a parent-child or tree-like structure
- Parents can have multiple children
- Children have field linking to parent
- "Expensive" to add data (need to go through tree)



Recap – Network DBs

- Like hierarchical DBs
- Multiple parents for children possible
- Difficult to change structure



Recap – Relational DBs

- Tables (rows and columns)
- Relations between tables/rows with keys

ID (PK)	OwnerID (FK)	Name	Race	Breed
P001	O001	Snoopy	DOG	Beagle
P002	O002	Garfield	CAT	Persian

ID (PK)	Name	Contact
O001	Charlie Brown	charlie.brown@example.com
O002	Jon Arbuckle	jon.arbuckle@example.com

Recap – Non-Relational DBs

- Not modelled in a tabular form
- Types
  - Document store
    - Pairs of key and document (complex data structure)
    - E.g., MongoDB, ElasticSearch
  - Key-value store
    - E.g., Redis
  - Wide-column store
    - Names and format of column can vary in rows
  - Graph store
    - E.g., Amazon Neptune, ArangoDB, Neo4j

- Further reading: https://insightsoftware.com/blog/whats-the-difference-relational-vs-non-relational-databases/

Managed Databases

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Definitions

*The definition of a managed database is a database with storage, data, and compute services that is **managed** and **maintained** by a **third-party** provider instead of by an organization's IT staff.*

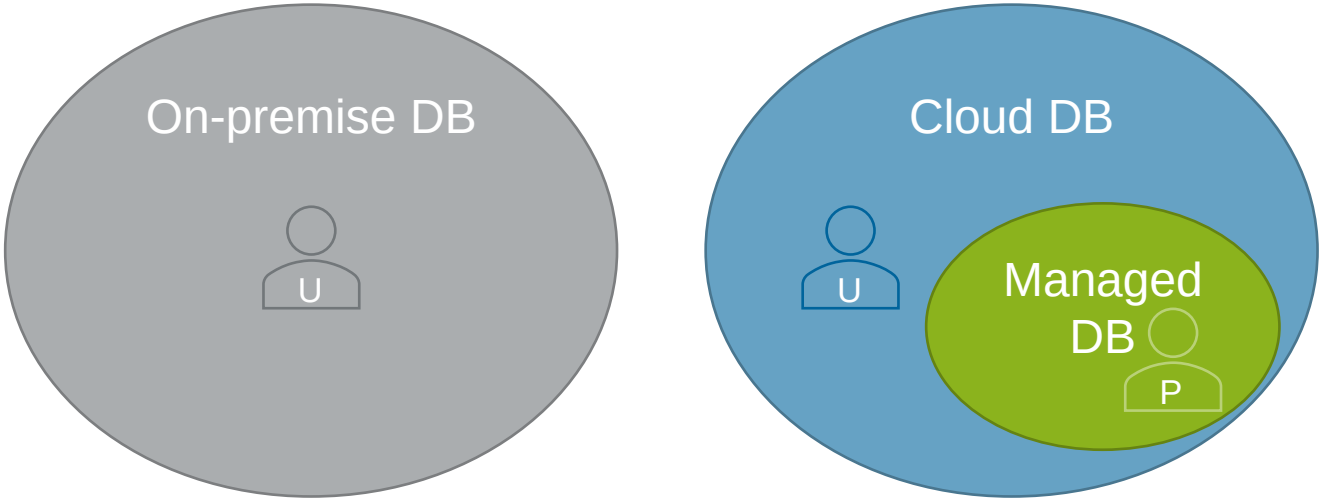
- Oracle (https://www.oracle.com/autonomous-database/what-is-managed-database/)

*A managed database is a cloud computing service in which the end user **pays** a cloud service provider for **access** to a database. Unlike a typical database, **users don't** have to **set up or maintain** a managed database on their own; rather, it's the **provider**'s responsibility to **oversee** the database's **infrastructure**.*

- Mark Drake (https://www.digitalocean.com/community/tutorials/understanding-managed-databases)

Differentiation

- Cloud DB ≠ Managed DB
- **NOT** every Cloud DB is a (fully) Managed DB
- It comes down to **who** manages the DB, User (U) or provider (P).



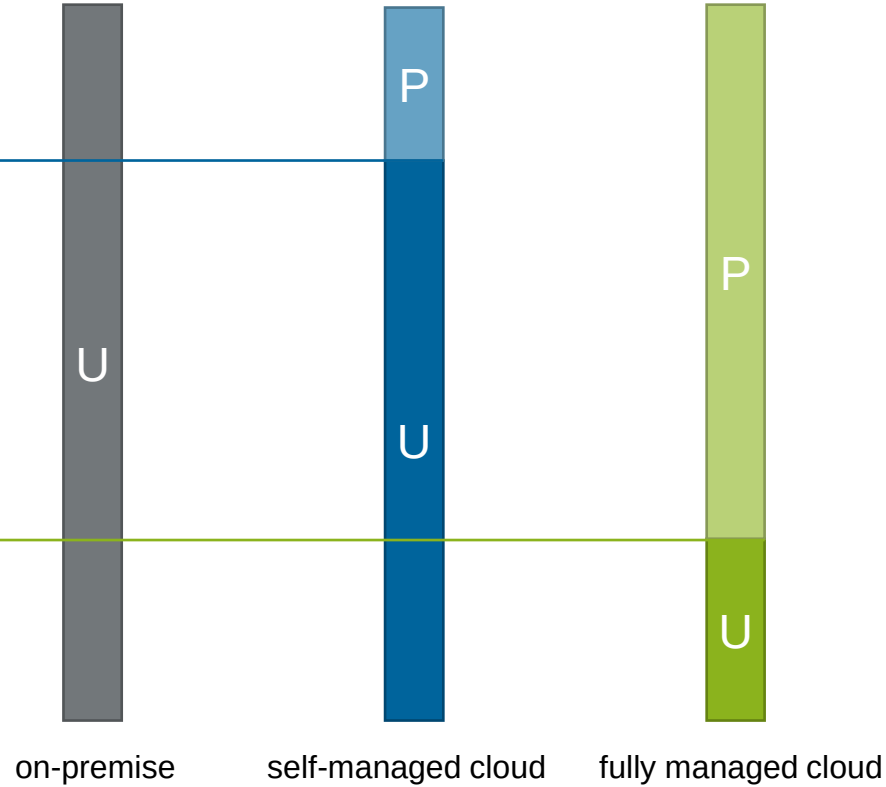
Useful terms

- Durability
  - After a transaction the data is permanently saved, even after system failure
  - Can be achieved by logging transactions, to replicate after failure
- Reliability
  - Consistent performance of DB
- Redundancy
  - Multiple copies of the same data is stored
- Availability
  - Up-time in which data can be reached

- Failover
  - When one DB fails, switch to other replica (redundancy)
  - Increases availability
- Read replica
  - Read-only instance of DB
  - Increases performance

Responsibilities

- Hardware
- Physical access security
- Backups
- Scaling
- Monitoring
- Maintenance/Patching
- Best practices
- Schema design
- Queries



U…User (Company that needs/uses DB)
P…Provider (Company providing DB e.g., Amazon AWS, Microsoft Azure, Google Cloud)

Advantages - Cloud

- Easy scaling
  - No need to wait for new hardware to arrive
  - Possibility of auto-scaling (based on API metrics)
- High availability
  - Redundancy: Failover/Backup in another location
- Lowering costs
  - Hardware
  - Security (e.g., physical access control)
  - Failover/Backup in a second location
  - Only pay what you use (“pay-as-you-go”)
- Increased security
  - In compliance with industry standards

Advantages - fully managed

- Lowering costs
  - Outsourced responsibilities
- Increased reliability
- Less responsibility
  - Maintenance
    - Optimization and trouble shooting
  - Patching and minor upgrade (no downtime)
    - Keeping OS and tools up-to-date
  - Automated backups and failover

Disadvantages

- Data Privacy and Security
- Responsibilities
  - Certification training of personnel is needed to use the services.
- Dependency (Vendor lock-in)
  - E.g., DB provider is sold, goes bankrupt
  - E.g., DB provider decides to no longer keep a feature you need

- Less control
  - E.g., when will which new security feature be patched

Examples of fully managed DBs

- AWS
  - Amazon RDS
  - Amazon DocumentDB
  - Amazon Keyspaces
  - Amazon ElastiCache
- Azure
  - Azure SQL
  - Azure Cosmos DB
  - Azure Database for PostgreSQL
  - Azure Database for MySQL
  - Azure Database for MariaDB
  - Azure Cache for Redis
  - Azure Managed Instance for Apache Cassandra
- GCP
  - Cloud SQL
  - Cloud Spanner
  - MongoDB Atlas
  - BigQuery
  - Cloud Bigtable
  - Cloud Firestore
  - Firebase Realtime Database
  - Cloud Memorystore

Comparison: https://acloudguru.com/blog/engineering/aws-vs-azure-vs-gcp-cloud-comparison-databases

Amazon RDS

- Relational Database Service

Source: https://aws.amazon.com/rds/

- Further reading & image source: https://aws.amazon.com/rds/features/

Amazon Document DB

- JSON data management at scale
- MongoDB compatibility
- 99.99% durability
- 6 replicas over in up to 5 AWS regions

- Further reading: https://aws.amazon.com/documentdb/features/

Amazon Keyspaces

- Apache Cassandra
- Serverless
- 99.99% availability
- 3 replicas across multiple Availability Zones (AZs)

- Further reading: https://aws.amazon.com/keyspaces/features/

Amazon ElastiCache

- Caching service
- Primary data store without durability
- Redis and Memcached compatibility

Source: https://aws.amazon.com/elasticache/

- Further reading: https://aws.amazon.com/elasticache/features

DB Migration

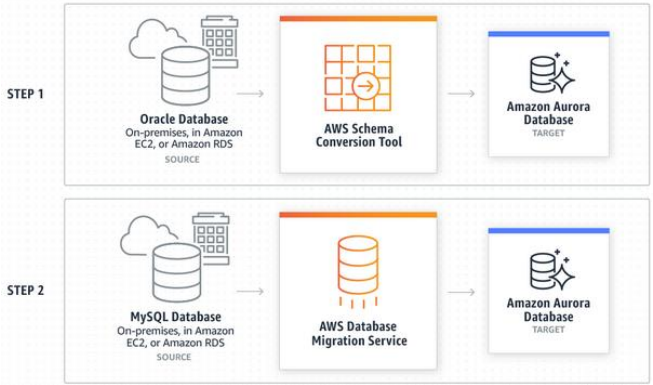
- Services available at AWS, Azure and GCP
- Use cases
  - Homogeneous migration
    - Source engine = target engine
      - e.g., PostgreSQL to PostgreSQL
  - Heterogeneous migration
    - Source engine ≠ target engine
      - e.g., Oracle to PostgreSQL
  - Database for development & testing purposes
  - DB consolidation
    - “merge” multiple DBs into one
  - Continuous replication within cloud or from cloud to on premise DB

Migration - AWS

- Homogeneous

- Heterogeneous

Source: https://aws.amazon.com/dms/



Videos

- Cloud Guru – Databases: https://youtu.be/ZbltenDTupQ
- AWS - Database Migration: https://youtu.be/ouia1Sc5QGo

References and Resources

- https://aws.amazon.com/developer/
- https://azure.microsoft.com/en-us/solutions/
- https://cloud.google.com/solutions
- https://cloud.netapp.com/blog/gcp-cvo-blg-google-cloud-database-the-right-service-for-your-workloads
- https://acloudguru.com/blog/engineering/aws-vs-azure-vs-gcp-cloud-comparison-databases
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- https://docs.aws.amazon.com/whitepapers/latest/logical-separation/encrypting-data-at-rest-and--in-transit.html
- https://aws.amazon.com/dms/
- https://insightsoftware.com/blog/whats-the-difference-relational-vs-non-relational-databases/